Paper D-04: Epigenetics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper develops a falsifiable CDFD/AFL model of Epigenetics. It maps developmental signals, environmental exposures, hormones, and metabolism to driving flux Φ, chromatin access, transcriptional capacity, repair, and cellular context to active constraint C, remodeling, feedback, compensation, and lineage-specific regulation to responsiveness S, and epigenetic marks, lineage, exposure history, and developmental timing to structural memory M_s. The target outcome is gene expression, phenotype, persistence, and reversibility, tested against multi-omics, lineage tracing, perturbation, exposure cohorts, and longitudinal phenotypes. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine

[5]. Part IV [6], DOI `https://doi.org/10.5281/zenodo.20395901`, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \to \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

The genome is not a static library; it is a bustling factory floor. RNA polymerases continuously scan the DNA, transcribing genetic code into the mRNA that builds the organism. However, not all parts of the factory floor are accessible. The cell physically tightly coils segments of DNA around histone proteins, locking them away from the transcriptional machinery.

In CDFD, this physical coiling is the quintessential mechanism of Adaptive Flux Limitation (AFL) at the molecular level.

3 The Genomic Adaptive Surface

We apply the Universal Adaptive Ratio to the epigenetic landscape:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

For a specific genetic locus:

- **Flux (Φ)**: The rate of transcription, measured by the volume of mRNA produced and the subsequent protein synthesis cascade.
- **Constraint (C)**: Chromatin compaction. Heterochromatin represents a high constraint state where genes are silenced; euchromatin represents a low constraint state where genes are freely accessible.
- **Responsiveness (S)**: The availability of transcription factors, RNA polymerase, and cellular ATP required to drive the expression process.
- **Memory (M_s)**: Epigenetic marks (e.g., DNA methylation and histone acetylation). These biochemical tags act as chronic memory locks. They ensure that a cell "remembers"

its constraint topology even after it divides, maintaining cellular identity (e.g., keeping a skin cell a skin cell).

3.1 Trauma, Adaptation, and Transgenerational Inheritance

During normal cellular differentiation, the adaptive ratio Ψ_s is carefully modulated to create distinct tissue types.

However, if the organism experiences severe environmental trauma (e.g., starvation, intense stress, or chronic inflammation), the organism must adapt to survive. The system aggressively alters its epigenetic memory (M_s). It may heavily methylate (lock down) certain metabolic genes to conserve energy, drastically increasing the local constraint C and dropping the transcription flux Φ.

Because these epigenetic marks constitute structural memory (M_s), they are incredibly persistent. The system locks itself into a defensive, highly constrained topology ($\Psi_s \ll 1$). Remarkably, this memory can be passed down to offspring. The progeny inherit the chronic constraints, leaving them predisposed to metabolic disorders (like obesity or anxiety) even in safe, resource-rich environments, because their genetic surface is pathologically over-constrained by their ancestral memory.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Major Universal Upgrade: Mechanism, Scale, and Tests

5.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Epigenetics, the proposed drive is developmental signals, environmental exposures, hormones, and metabolism. The active limitation is chromatin access, transcriptional capacity, repair, and cellular context. Responsiveness is carried by remodeling, feedback, compensation, and lineage-specific regulation, while retained state is represented by epigenetic marks, lineage, exposure history, and developmental timing. The outcome to explain is gene expression, phenotype, persistence, and reversibility.

Table 1: Operational CDFD/AFL map for Paper D-04.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	developmental signals, environmental exposures, hormones, and metabolism	What rate, load, gradient, or demand enters the focal system?
C	chromatin access, transcriptional capacity, repair, and cellular context	Which measured bottleneck limits transfer, function, or recovery?
S	remodeling, feedback, compensation, and lineage-specific regulation	Which process changes effective capacity or reroutes the load?
M_s	epigenetic marks, lineage, exposure history, and developmental timing	Which retained state makes equal present loads produce different futures?
Y	gene expression, phenotype, persistence, and reversibility	Which outcome is predicted out of sample, with uncertainty?

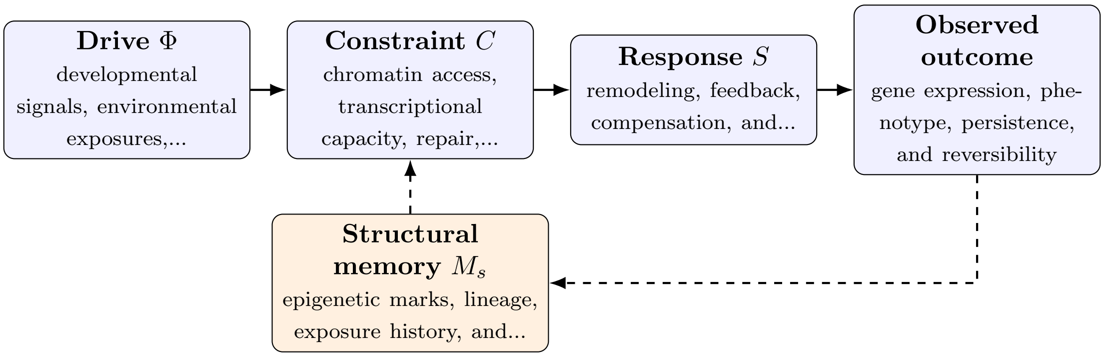

Figure 1: Paper D-04 observable CDFD/AFL architecture for Epigenetics. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

5.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \tag{2}$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \tag{3}$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \tag{4}$$

$$\Psi_s(t) = \frac{\widehat{\Phi}(t)}{\epsilon + \widehat{C}(t)}\,\widehat{S}(t)\,[1 + \omega\widehat{M}_s(t)]. \tag{5}$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in developmental signals, environmental exposures, hormones, and metabolism arrives faster than remodeling, feedback, compensation, and lineage-specific regulation can compensate. This raises or redistributes chromatin access, transcriptional capacity, repair, and cellular context. If the load persists, the retained state encoded by epigenetic marks, lineage, exposure history, and developmental timing changes the next response, so a later event of equal magnitude can produce a different trajectory in gene expression, phenotype, persistence, and reversibility. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

5.3 Cross-Scale Universality Without Mechanistic Erasure

For the domain applications family, the universal comparison is a disciplined translation test across systems with very different material mechanisms. A valid mapping preserves the baseline science of the domain, declares the scale of every variable, and earns its place through a discriminating prediction rather than resemblance alone. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is minutes to generations; loci to organisms. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

5.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper D-04. Each rung can fail independently.

Test	**Design**	**Discriminating result**
Proxy audit	Measure multi-omics, lineage tracing, perturbation, exposure cohorts, and longitudinal phenotypes and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a controlled exposure followed by removal or reprogramming while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: past exposure does not alter response after current molecular state is controlled.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ, separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

5.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from multi-omics, lineage tracing, perturbation, exposure cohorts, and longitudinal phenotypes and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of gene expression, phenotype, persistence, and reversibility. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

6 Conclusion

Epigenetic state is a concrete molecular form of biological memory whose mechanisms must be studied on their own terms. CDFD supplies a high-level language for testing history-dependent constraint and response, but it does not erase the mechanistic gap between chromatin regulation, trauma, and economic systems.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

7 Reference Layer

References

[1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the 1/f noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: `https://doi.org/10.1103/PhysRevLett.59.381`.

[2] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973. DOI: `https://doi.org/10.1146/annurev.es.04.110173.000245`.

[3] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026. DOI: `https://doi.org/10.5281/zenodo.20250821`.

[4] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026. DOI: `https://doi.org/10.5281/zenodo.20264779`.

[5] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026. DOI: `https://doi.org/10.5281/zenodo.20344695`.

[6] Steve Bico Mujjabi. Cdfd part iv: Universal adaptive flux limitation - constraint-driven flux dynamics, 2026. DOI: `https://doi.org/10.5281/zenodo.20395901`.

[7] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026. DOI: `https://doi.org/10.5281/zenodo.20343160`.

[8] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: `https://doi.org/10.1002/j.1538-7305.1948.tb01338.x`.

[9] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: `https://doi.org/10.1098/rstb.1952.0012`.